Eg:- Direction of the velocity means direction of the motion.

① Find the velocity, acceleration and speed at given time 't' of a particle moving along the given curve

$x = 1+3t, \quad y = 3-4t, \quad z = 7+3t.$

Ans:-

The position vector $\vec{r}(t) = x\hat{i} + y\hat{j} + z\hat{k}$

ie, $\vec{r}(t) = (1+3t)\hat{i} + (3-4t)\hat{j} + (7+3t)\hat{k}$

Velocity $= \vec{v}(t) = \frac{d\vec{r}}{dt}$

$= 3\hat{i} - 4\hat{j} + 3\hat{k}$

Acceleration $= \vec{a}(t) = \frac{d\vec{v}}{dt} = 0$

Speed $= \|\vec{v}(t)\| = \sqrt{3^2+(-4)^2+3^2} = \sqrt{34}$

② A particle moves along a circular path in such a way that its x and y coordinates at time t are $x = 2\cos t, \quad y = 2\sin t.$

① Find the instantaneous velocity, acceleration and speed at time t.

② Find the position vector, velocity vector and acceleration vector at $t = \pi/4$

③ Show that at each instant the acceleration vector is perpendicular to the velocity vector.

Ans:-

① The position vector $\vec{r}(t) = x\hat{i} + y\hat{j} \cancel{+ z\hat{k}}$

$\therefore \vec{r}(t) = 2\cos t\,\hat{i} + 2\sin t\,\hat{j}$

Velocity $= \vec{v}(t) = \dfrac{d\vec{r}}{dt} = -2\sin t\,\hat{i} + 2\cos t\,\hat{j}$

Acceleration $= \vec{a}(t) = \dfrac{d\vec{v}}{dt} = -2\cos t\,\hat{i} - 2\sin t\,\hat{j}$

Speed $= \|\vec{v}(t)\| = \sqrt{(-2\sin t)^2 + (2\cos t)^2}$

$= \underline{2}.$

② $\vec{r}(t) = 2\cos t\,\hat{i} + 2\sin t\,\hat{j}$

$\vec{r}(\pi/4) = 2\cos(\pi/4)\,\hat{i} + 2\sin(\pi/4)\,\hat{j}$

$= \dfrac{2}{\sqrt{2}}\,\hat{i} + \dfrac{2}{\sqrt{2}}\,\hat{j} = \underline{\sqrt{2}\,\hat{i} + \sqrt{2}\,\hat{j}}$

$\vec{v}(\pi/4) = -2\sin(\pi/4)\,\hat{i} + 2\cos(\pi/4)\,\hat{j}$

$= \underline{-\sqrt{2}\,\hat{i} + \sqrt{2}\,\hat{j}}$

$$\vec{a}(\pi/4) = -2\cos(\pi/4)\,\hat{i} - 2\sin(\pi/4)\,\hat{j}$$

$$= -\sqrt{2}\,\hat{i} - \sqrt{2}\,\hat{j}$$

③ $\vec{v}(t) \cdot \vec{a}(t) = (-2\sin t\,\hat{i} + 2\cos t\,\hat{j}) \cdot (-2\cos t\,\hat{i} - 2\sin t\,\hat{j})$

$= 4\cos t\sin t - 4\cos t\sin t$

$= 0.$

$\Rightarrow \vec{v}(t)$ and $\vec{a}(t)$ are perpendicular

③ A particle moves through 3-space in such a way that its velocity is $\vec{V}(t) = \hat{i} + t\hat{j} + t^2\hat{k}$. Find the co-ordinates of the particle at time $t=1$ given that the particle is at the point $(-1, 2, 4)$ at time $t=0$.

Ans:-

$$\vec{v}(t) = \frac{d\vec{r}}{dt}$$

$$\Rightarrow \vec{r}(t) = \int \vec{v}(t)\,dt$$

$$= \int (\hat{i} + t\hat{j} + t^2\hat{k})\,dt$$

$$= t\hat{i} + \frac{t^2}{2}\hat{j} + \frac{t^3}{3}\hat{k} + C \longrightarrow ①$$

Given that at time $t=0$, $\vec{r}(0)$ is at $(-1, 2, 4)$.

$\therefore \vec{r}(0) = -\hat{i} + 2\hat{j} + 4\hat{k}$

$\therefore (1) \Rightarrow c = -\hat{i} + 2\hat{j} + 4\hat{k}$

$\therefore \vec{r}(t) = \left(t\hat{i} + \frac{t^2}{2}\hat{j} + \frac{t^3}{3}\hat{k}\right) + \left(-\hat{i} + 2\hat{j} + 4\hat{k}\right)$

$= (t-1)\hat{i} + \left(\frac{t^2}{2} + 2\right)\hat{j} + \left(\frac{t^3}{3} + 4\right)\hat{k}$

$\therefore$ At time $t=1$,

$\vec{r}(t) = \vec{r}(1) = 0\hat{i} + \left(\frac{1}{2} + 2\right)\hat{j} + \left(\frac{1}{3} + 4\right)\hat{k}$

$= 0\hat{i} + \frac{5}{2}\hat{j} + \frac{13}{3}\hat{k}$

$\therefore$ Co-ordinates at time $t=1$ are $\left(0, \frac{5}{2}, \frac{13}{3}\right)$

(4) Find the velocity and position vectors of the particle, if the acceleration vector $\vec{a}(t) = (\sin t)\hat{i} + (\cos t)\hat{j} + e^t\hat{k}$, $\vec{v}(0) = \hat{k}$

$\vec{r}(0) = -\hat{i} + \hat{k}$.

Ans:-

$\vec{a}(t) = (\sin t)\hat{i} + (\cos t)\hat{j} + (e^t)\hat{k}$

$\vec{a}(t) = \frac{d\vec{v}}{dt}$

$\therefore \vec{v}(t) = \int \vec{a}(t)\,dt = \int (\sin t)\hat{i} + (\cos t)\hat{j} + (e^t)\hat{k}\,dt$

$\therefore \vec{v} = \int \vec{a}(t)\,dt$

$$= (-\cos t)\hat{i} + (\sin t)\hat{j} + e^t\hat{k} + c_1 \longrightarrow ①$$

Given that $\vec{v}(0) = \hat{k}$.

$$① \Rightarrow \vec{v}(0) = -\hat{i} + 0\hat{j} + \hat{k} + c_1 = \hat{k}$$

$$\Rightarrow \boxed{c_1 = \hat{i}}$$

$$\therefore \vec{v}(t) = (-\cos t)\hat{i} + (\sin t)\hat{j} + e^t\hat{k} + \hat{i}$$

Now $\vec{r}(t) = \int \vec{v}(t)\,dt$ $\quad \boxed{\vec{v}(t) = \frac{d\vec{r}}{dt}}$

$$= \int \left[\left((-\cos t)\hat{i} + (\sin t)\hat{j} + e^k\hat{k} \right) + \hat{i} \right] dt$$

$$= (-\sin t)\hat{i} - (\cos t)\hat{j} + e^t\hat{k} + t\,\hat{i} + c_2 \longrightarrow ②$$

Given that $\vec{r}(0) = -\hat{j} + \hat{k}$

$$\therefore ② \Rightarrow -\hat{i} + \hat{k} = 0\hat{i} - \hat{j} + \hat{k} + 0\hat{i} + c_2$$

$$\Rightarrow \boxed{c_2 = -\hat{i} + \hat{j}}$$

$$\therefore \vec{r}(t) = \left[(-\sin t)\hat{i} + (-\cos t)\hat{j} + e^t\hat{k} + t\,\hat{i} \right] + -\hat{i} + \hat{j}$$

$$\therefore \vec{r}(t) = (t - \sin t - 1)\hat{i} + (1 - \cos t)\hat{j} + e^t\hat{k}$$

I Find the velocity, acceleration and speed of a particle moving along the following curves.

① $\vec{r}(t) = (2\cos t)\hat{i} + (2\sin t)\hat{j} + t\hat{k}$ at $t = \pi/2$.

② $\vec{r}(t) = (e^t \sin t)\hat{i} + (e^t \cos t)\hat{j} + t\hat{k}$, at $t = \pi$

③ $\vec{r}(t) = (3\cos t)\hat{i} + (3\sin t)\hat{j}$, at $t = \pi/3$

Homework Answer [illegible]

Qn. Find the velocity, acceleration and speed of $\vec{r}(t) = (e^t \sin t)\hat{i} + (e^t \cos t)\hat{j} + t\hat{k}$ at $t=\pi$

Ans:-

$$\vec{v}(t) = \frac{d\vec{r}}{dt} = \frac{d}{dt}\left[(e^t \sin t)\hat{i} + (e^t \cos t)\hat{j} + t\hat{k}\right]$$

$$= (e^t \cos t + e^t \sin t)\hat{i} + (e^t(-\sin t) + e^t \cos t)\hat{j} + \hat{k}$$

$\left[\text{By product rule}\right]$

$$= \left[e^t(\cos t + \sin t)\right]\hat{i} + \left[e^t(\cos t - \sin t)\right]\hat{j} + \hat{k}$$

$\vec{v}(t)$ at $t=\pi$ is

$$\vec{v}(\pi) = \left[e^{\pi}(\cos\pi + \sin\pi)\right]\hat{i} + \left[e^{\pi}(\cos\pi - \sin\pi)\right]\hat{j} + \hat{k}$$

$$= e^{\pi}(-1)\hat{i} + e^{\pi}(-1)\hat{j} + \hat{k}$$

$$= -e^{-\pi}\left[\hat{i} + \hat{j}\right] + \hat{k}$$

$\cos\pi = -1$

$\sin\pi = 0.$

$$\vec{a}(t) = \frac{d\vec{v}}{dt} = \frac{d}{dt}\left(e^t(\cos t + \sin t)\hat{i} + (e^t(\cos t - \sin t))\hat{j} + \hat{k}\right)$$

$$= \left[e^t\left[-\sin t + \cos t\right] + e^t(\cos t + \sin t)\right]\hat{i} + \left[e^t(-\sin t - \cos t) + e^t(\cos t - \sin t)\right]\hat{j} + 0$$

$$= (2e^t \cos t)\hat{i} + (-2e^t \sin t)\hat{j}$$

$$\vec{a}(\pi) = (2e^{\pi}\cos\pi)\hat{i} + (-2e^{\pi}\sin\pi)\hat{j} = -2e^{\pi}\hat{i}$$

Speed $= \|\vec{v}(t)\|$

$$= \sqrt{[e^t(\cos t + \sin t)]^2 + [e^t(\cos t - \sin t)]^2 + 1}$$

$$= \sqrt{e^{2t}(\cos^2 t + 2\cos t \sin t + \sin^2 t) + e^{2t}(\cos^2 t - 2\cos t \sin t + \sin^2 t) + 1}$$

$(e^t)^2 = e^{2t}$

$$= \sqrt{e^{2t}(2\cos^2 t + 2\sin^2 t) + 1}$$

$$= \sqrt{e^{2t}\, 2(\cos^2 t + \sin^2 t) + 1}$$

$$= \sqrt{2e^{2t} + 1}$$

Speed at $t = \pi$ is $\underline{\underline{\sqrt{2e^{2\pi} + 1}}}$

Problems on minimum and maximum speed

(4) The motion of the particle is described by the position vector $\vec{r} = (t-t^2)\hat{i} - t^2\hat{j}$. Find the minimum speed of the particle and its location when it has this speed.

Ans:-

$\vec{r} = (t-t^2)\hat{i} - t^2\hat{j}$

Speed $= \|\vec{v}(t)\|$

$\vec{v}(t) = \dfrac{d\vec{r}}{dt} = \dfrac{d}{dt}\left[(t-t^2)\hat{i} - t^2\hat{j}\right]$

$= (1-2t)\hat{i} - 2t\hat{j}$

$\|\vec{v}(t)\| = \sqrt{(1-2t)^2 + (-2t)^2}$

$= \sqrt{1-4t+4t^2+4t^2}$

$= \sqrt{1-4t+8t^2}$

For max or min

$f'(x) = 0$ for values of x.

If $f''(x) > 0$, $f(x)$ has minima

If $f''(x) < 0$, $f(x)$ has maxima

at those points of x.

For checking Maxima or Minima

let $f(t) = 1-4t+8t^2$

$f'(t) = -4+16t = 0$

$\Rightarrow 16t = 4 \Rightarrow t = \frac{4}{16} = 1/4$

Now $f''(t) = 16 > 0$ for all t

$\Rightarrow f(t)$ has a minimum at $t = 1/4$

Hence minimum speed is at $t = \frac{1}{4}$

$\therefore$ Minimum speed $= \|\vec{v}(t)\|$ at $t = 1/4$

$$= \sqrt{1 - 4 \times \frac{1}{4} + 8 \times \frac{1}{4^2}}$$

$$= \sqrt{1 - 1 + \frac{1}{2}}$$

$$= \sqrt{1/2}$$

Location of the particle when it has minimum speed is given by $\vec{r}(1/4)$.

$$\vec{r}(t) = (t - t^2)\hat{i} - t^2\hat{j}$$

$$\vec{r}(1/4) = \left(\frac{1}{4} - \frac{1}{16}\right)\hat{i} - \frac{1}{16}\hat{j}$$

$$= \frac{3}{16}\hat{i} - \frac{1}{12}\hat{j}$$

⑤ The position vector of a particle moving in 2-space is given by

$$\vec{r}(t) = (\sin 3t)\hat{i} - (2\cos 3t)\hat{j}, \quad 0 \le t \le 2\pi/3$$

Find the maximum and minimum speed of the particle.

Ans:-

Speed of the particle $= \|\vec{v}(t)\|$

$$\vec{r}(t) = (\sin 3t)\hat{i} - 2\cos 3t\,\hat{j}$$

$$\vec{v}(t) = \frac{d\vec{r}}{dt} = \frac{d}{dt}(\sin 3t)\hat{i} - (2\cos 3t)\hat{j}$$

$$= (3\cos 3t)\hat{i} + (6\sin 3t)\hat{j}$$

$$\|\vec{v}(t)\| = \sqrt{(3\cos 3t)^2 + (6\sin 3t)^2}$$

$$= \sqrt{9\cos^2 3t + 36\sin^2 3t}$$

$$= \sqrt{9(1-\sin^2 3t) + 36\sin^2 3t}$$

$$= \sqrt{9 - 9\sin^2 3t + 36\sin^2 3t}$$

$$= \sqrt{9 + 27\sin^2 3t}$$

is maximum when $\sin^2 3t$ is maximum (ie $\sin^2 3t = 1$) and is minimum when $\sin^2 3t$ is minimum (ie $\sin^2 3t = 0$)

Maximum speed of the particle is when $\sin^2 3t = 1$

$$\therefore \|\vec{v}(t)\| = \sqrt{9+27} = \sqrt{36} = \underline{\underline{6}}$$

Minimum speed of the particle is when $\sin^2 3t = 0$

$$\therefore \|\vec{v}(t)\| = \sqrt{9+0} = \sqrt{9} = \underline{3}$$

Also $\sin 3t = \pm 1$ occurs first when $\boxed{t = \pi/6}$ [since $\sin\frac{\pi}{2} = 1$]

And $\sin 3t = 0$ occurs first when $\boxed{t = 0}$

Homework

(1) The position vector of a particle is

$\vec{r} = 12\sqrt{t}\,\hat{i} + t^{3/2}\,\hat{j}, \quad t > 0.$

Find the minimum speed of the particle and the location when it has this speed.

(2) Find the maximum and minimum speed of the particle moving in a 3-space with position vector

$\vec{r}(t) = (3\cos 2t)\,\hat{i} + (5\sin 2t)\,\hat{j} + (4t)\,k$

① Suppose that a particle moves along a circular helix in 3-space so that its position vector at time t is

$\vec{r}(t) = (4\cos\pi t)\hat{i} + (4\sin\pi t)\hat{j} + t\hat{k}$.

Find the distance travelled and displacement of the particle during the time interval $1 \le t \le 5$.

Ans:-

Given that $\vec{r}(t) = (4\cos\pi t)\hat{i} + (4\sin\pi t)\hat{j} + t\hat{k}$

$\therefore \vec{v}(t) = \frac{d\vec{r}}{dt} = (-4\pi\sin\pi t)\hat{i} + (4\pi\cos\pi t)\hat{j} + \hat{k}$

$$\|\vec{v}(t)\| = \sqrt{(-4\pi \sin \pi t)^2 + (4\pi \cos \pi t)^2 + 1^2}$$

$$= \sqrt{16\pi^2 (\sin^2 \pi t + \cos^2 \pi t) + 1}$$

$$= \sqrt{16\pi^2 + 1}$$

$$\text{Distance travelled} = \int_{t_1}^{t_2} \|\vec{v}(t)\| \, dt$$

$$\boxed{= \int_1^5 \sqrt{16\pi^2+1} \, dt = \sqrt{16\pi^2+1} \int_1^5 dt}$$

$$= \sqrt{16\pi^2+1} \, [t]_1^5$$

$$= \sqrt{16\pi^2+1} \, (5-1) = \underline{\underline{4\sqrt{16\pi^2+1}}}$$

$$\text{Displacement} = \int_{t_1}^{t_2} \vec{v}(t) \, dt = \vec{r}(t_2) - \vec{r}(t_1)$$

$$\vec{r}(t_2) = \vec{r}(5) = (4 \cos 5\pi)\hat{i} + (4 \sin 5\pi)\hat{j} + 5\hat{k}$$

$$\vec{r}(t_1) = \vec{r}(1) = (4 \cos \pi)\hat{i} + (4 \sin \pi)\hat{j} + \hat{k}$$

$$\vec{r}(t_2) - \vec{r}(t_1) = \vec{r}(5) - \vec{r}(1)$$

$$= (4 \cos 5\pi - 4 \cos \pi)\hat{i} + (4 \sin 5\pi - 4 \sin \pi)\hat{j} + (5-1)\hat{k}$$

$$= 4[-1 - -1]\hat{i} + 4[0-0]\hat{j} + 4\hat{k}$$

$$= \underline{\underline{4\hat{k}}}$$